Machine-Learning-Enabled Design for Molten Salt Reactors

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Big Data for NPPs Workshop, The Ohio State University Columbus, OH - Dec. 10-11, 2019

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Motivation/Aim for This Study

- To determine an optimized core design of for Molten Salt Reactors.
- To use a novel, robust, reliable, and accelerated method in the designing process.
- To make the reactor more efficient, compact, and safe
- **Aim:** To introduce an efficient, robust, and reliable design recommendation approach based on machine learning methods
- This talk presents preliminary results using AI/ML to find the optimized reactor parameters rather than traditional optimization methods such genetic algorithms.

Programs/Codes Used

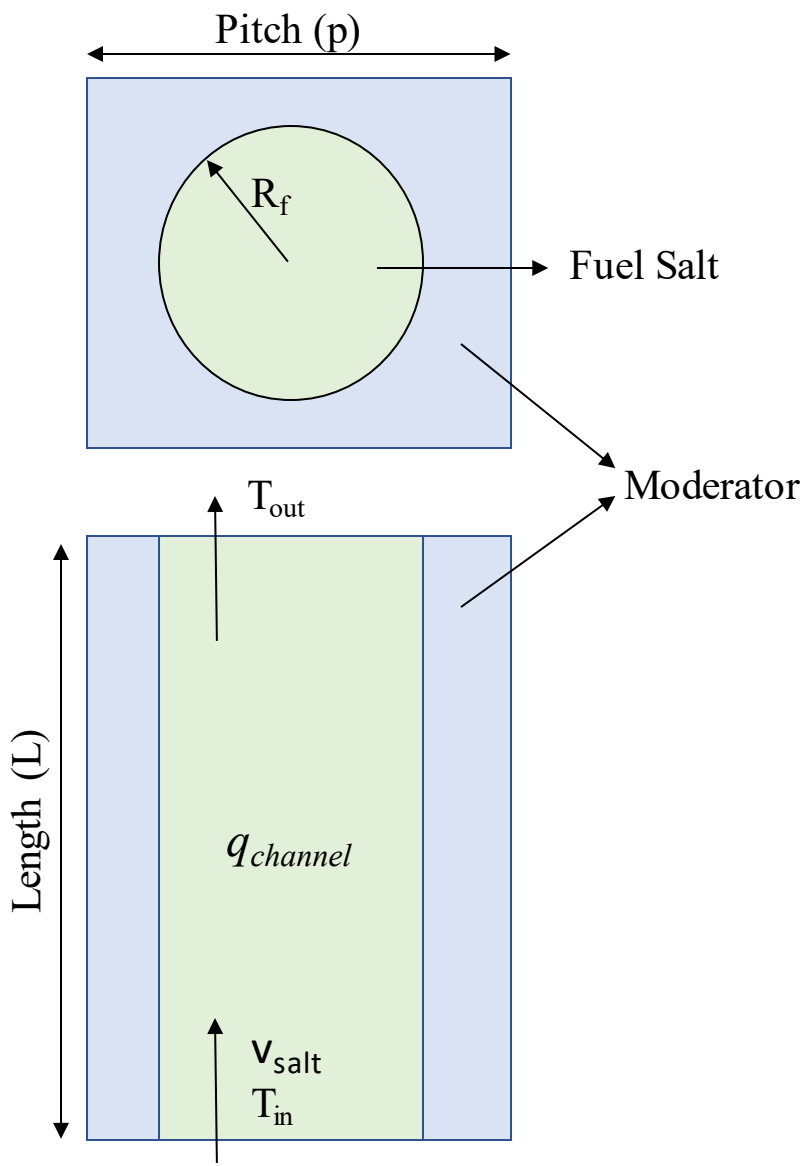
- Criticality calculations: Serpent 2 – Version 2.1.36
  - Computational standard deviation of $k_\infty < 100$ pcm
- Neutron data library: ENDF/B-VII.0
  - Serpent is not capable of using state-of-the-art ENDF libraries
- Programming: Python 3.7
- Sampling/Training: Raven Framework (INL)
  - Stable version (Nov 19) (https://raven.inl.gov/SitePages/Code.aspx)

Methods Used

- Reactor model: *Monte Carlo Technique (Serpent 2)*
- Target reactor design: *Molten Salt Reactor*
- Sampling methods: *Grid Sampling Approach*
- Training methods: ROMs
  a. sci-kit learn
     i. Neural Network
     ii. Decision Tree
     iii. K-Nearest Neighbors
- Classification: Data Mining
  - Ensemble
  - Cluster
- Method validation: single fuel channel with unit cell approximation

Method Validation

- Single fuel channel using unit cell approximation
- For square lattice cell geometry, 8 independent variables are under investigation for optimization.
- **Features:**
  1) *Fuel type*
  2) *Salt type, density and flow rate*
  3) *Enrichment (ε)*
  4) *Pitch (p) and channel radius (R_f)*
  5) *Moderator to fuel ratio*
  6) *Channel length*
  7) $q_{channel}$ *(or in terms of T_{in} and T_{out})*



Input/Target Parameters Selected

- **Inputs:**
  a. *Fuel types: U, U-Th, U-Pu*
     - *Th as Th-232*
     - *Pu as weapons-grade Pu*
  b. *Salt types: LiF-BeF2, NaF-BeF2, NaCl*
     (i) *LiF+BeF2 (72/16 mole %)+HM (12 mole %)* - MSFR/Mosart/Flibe
     *rho(T)=3.628-6.6e-4*T(°C)*[1] *for U-Th*
     (ii) *NaF+BeF2 (76/12 mole %)+HM (12 mole %)* - ThorCon
     *rho(T)=5.52-1.964e-3*T(°C)*[2] *for U*
     (iii) *NaCl (60 mole %)+HM (40 mole %)* - Terrapower/Elysium/Moltex
     *rho(T)=4.29-1.59e-3*T(°C)*[2] *for U*

[1] Ref. Jerden, J., 2019. Molten Salt Thermophysical Properties Database Development: 2019 Update. Chemical and Fuel Cycle Technologies Division, ANL/CFCT-19/6
[2] Ref. Janz, G.J., 1988. Thermodynamic and Transport Properties for Molten Salts: Correlation Equations for Critically Evaluated Density, Surface Tension, Electrical Conductance, and Viscosity Data. ThermoJournal of Physical and Chemical Reference Data 17 (2) 1–309.

Input/Target Parameters Selected

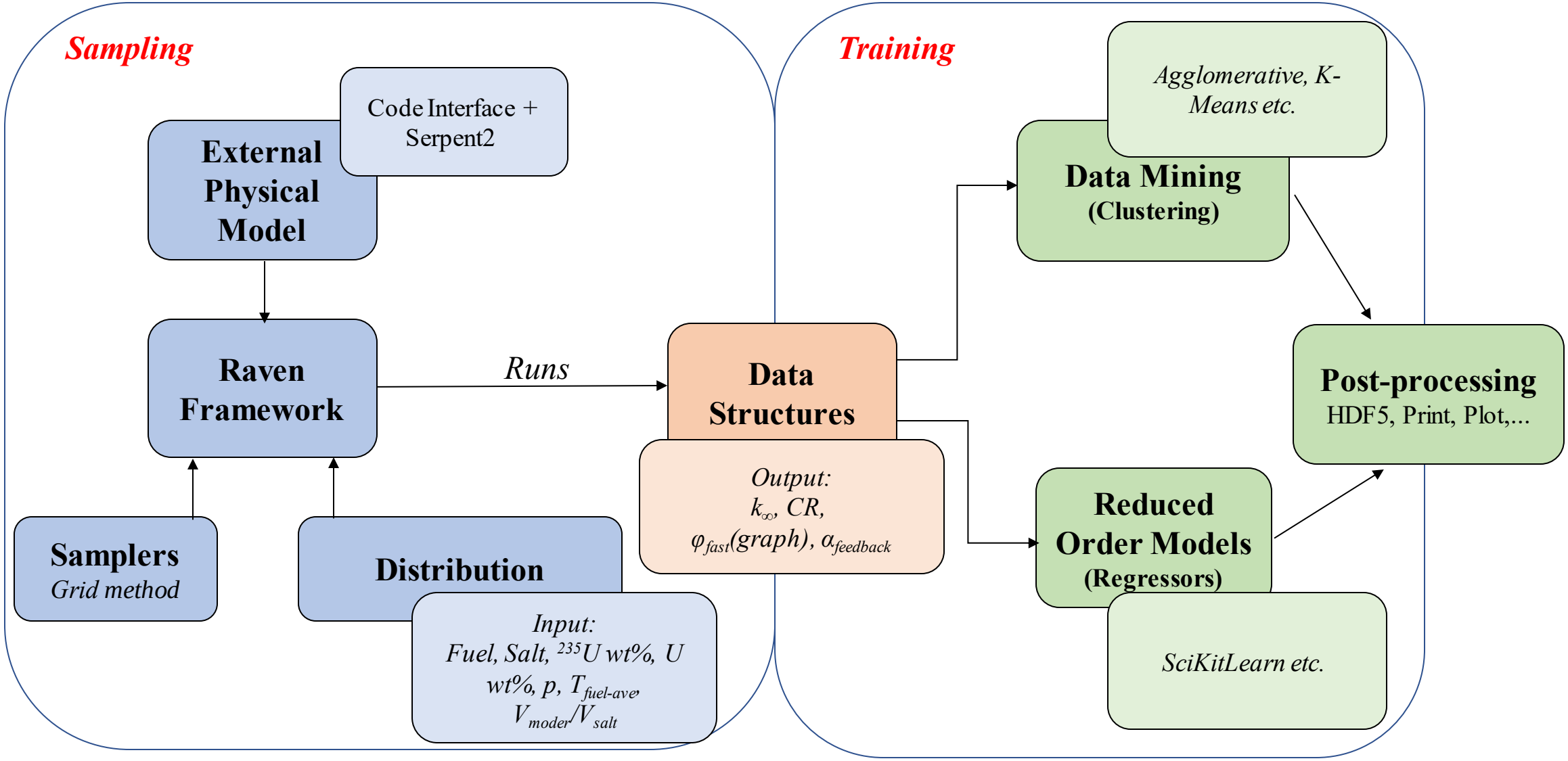
- **Inputs:**
  a. *^{235}U fissile content (ε): 1 to 20 wt. % in HM*
  b. *U content: 10 to 90 wt. % in HM*
     - *100 wt. % for U type*
  c. *Pitch (p): 1 to 16 cm*
     - *based on MSRE design*
  d. *V_{moder}/V_{salt}: 0.274 to 10.46*
     - *minimum value: 0.274 for $p=D_f$*
     - *maximum value: 10.46 for $p=3D_f$*
  e. *$T_{fuel\text{-}average}$: 900 to 1200 K*

- **Outputs:**
  a. *Infinite multiplication factor (k_∞)*
  b. *Conversion ratio (CR)*
  c. *Fast flux on moderator*
  d. *Total feedback coefficient ($\alpha_D+\alpha_V+...$)*

Assumptions/Approximations/Simplifications

- $T_{fuel\text{-}aver}$ instead of $q'_{channel}$ over a fuel channel length of 1m
- Only neutronic calculations are under consideration
  i. Thermal-hydraulics effects are ignored
  ii. Stationary fuel salt
- Base-line temperature 900 K for feedback calculations
  i. Temperature is the main driver for fuel salt density and Doppler feedback
- Calculations just for initial fuel loading

Computational Flowchart



Grid Sampling Approach

- The simplest exploration approach to explore an uncertain domain
- Construct an N-dimensional grid where each dimension is represented by one uncertain variable
- Equal distances
- Performs the sampling at each node of the grid
- Total number of samples: 31500

Variable	**Fuel Type**	**Salt Type**	**^{235}U Content**	**U content**	**Pitch**	**V_{moder}/V_{salt}**	**$T_{fuel\text{-}average}$**
Distribution	Discrete	Discrete	Continuous	Continuous	Continuous	Continuous	Continuous
Lower/Upper Values	U, U-Th, U-Pu	LiF-BeF2, NaF-BeF2, NaCl	1-20 wt.%	10-90 wt.%	1-16 cm	0.274-10.46	900-1200
Step Sizes	3	3	7	5	5	5	4

Data Mining

Options	KMeans	AgglomerativeClustering	DBScan
Classification	cluster	cluster	cluster
Solver	k-means++	ward	eps=0.2
# of Clusters	8	8	min_samples=20
Metric		euclidean	euclidean

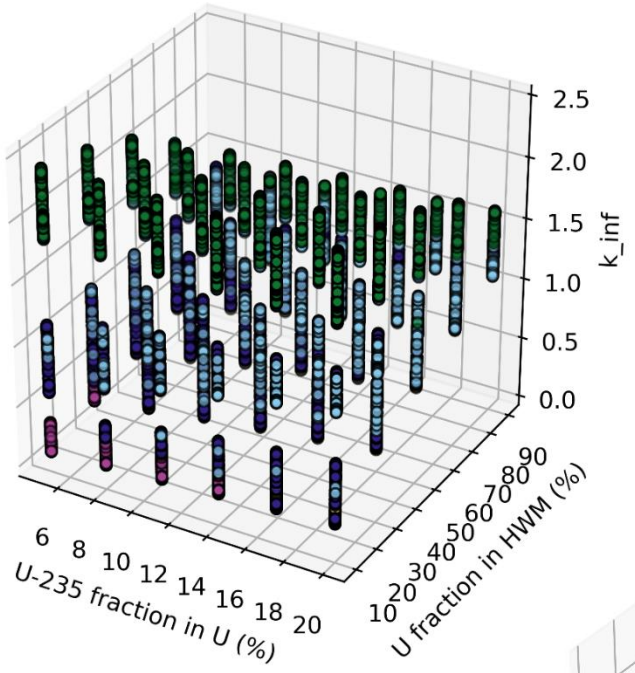
- Search for how the targets are to be clustered
- Identify the decision boundaries and classes
- Training data: Inputs (fuel type, ...) + Outputs (k_∞, ...)
- Those methods do not necessarily require target(s)
- Rest of the options are set to the default value

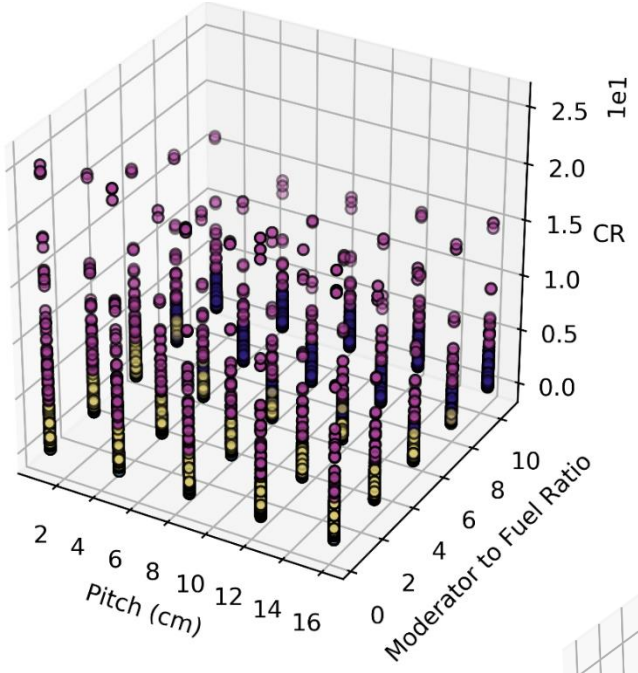
Reduced Order Methods

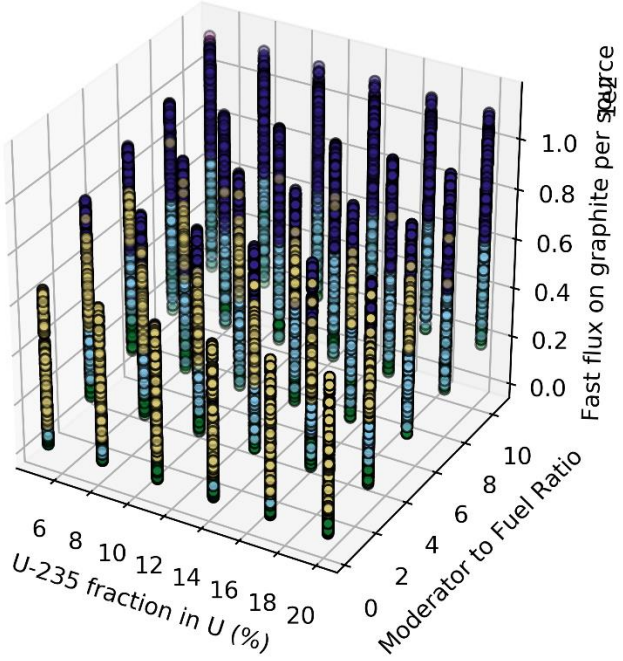
Methods	Class	Options
RandomForestRegressor	Ensemble	n_estimators=200
DecisionTreeRegressor	Decision Tree	-
KNeighborsRegressor	Neighbor Based	n_neighbors=20
MLPRegressor	Neural Network	max_iter=2000

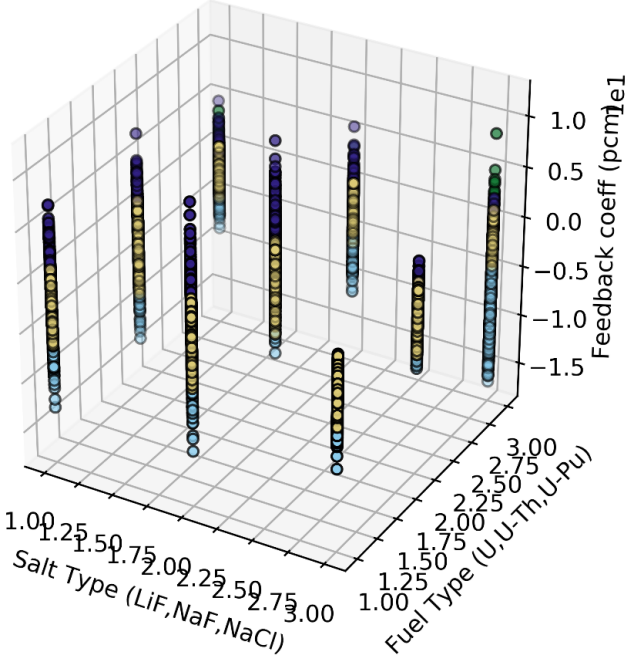
- To predict the target values
- Training data: Inputs (fuel type, ...)
- Target values: Outputs (k_∞, ...)
- Rest of the options are set to the default value
- Some regressors such as AdaBoost Regressor are incapable of training our datasets as they do not support multi-output feature

Results: Raw Data Distribution

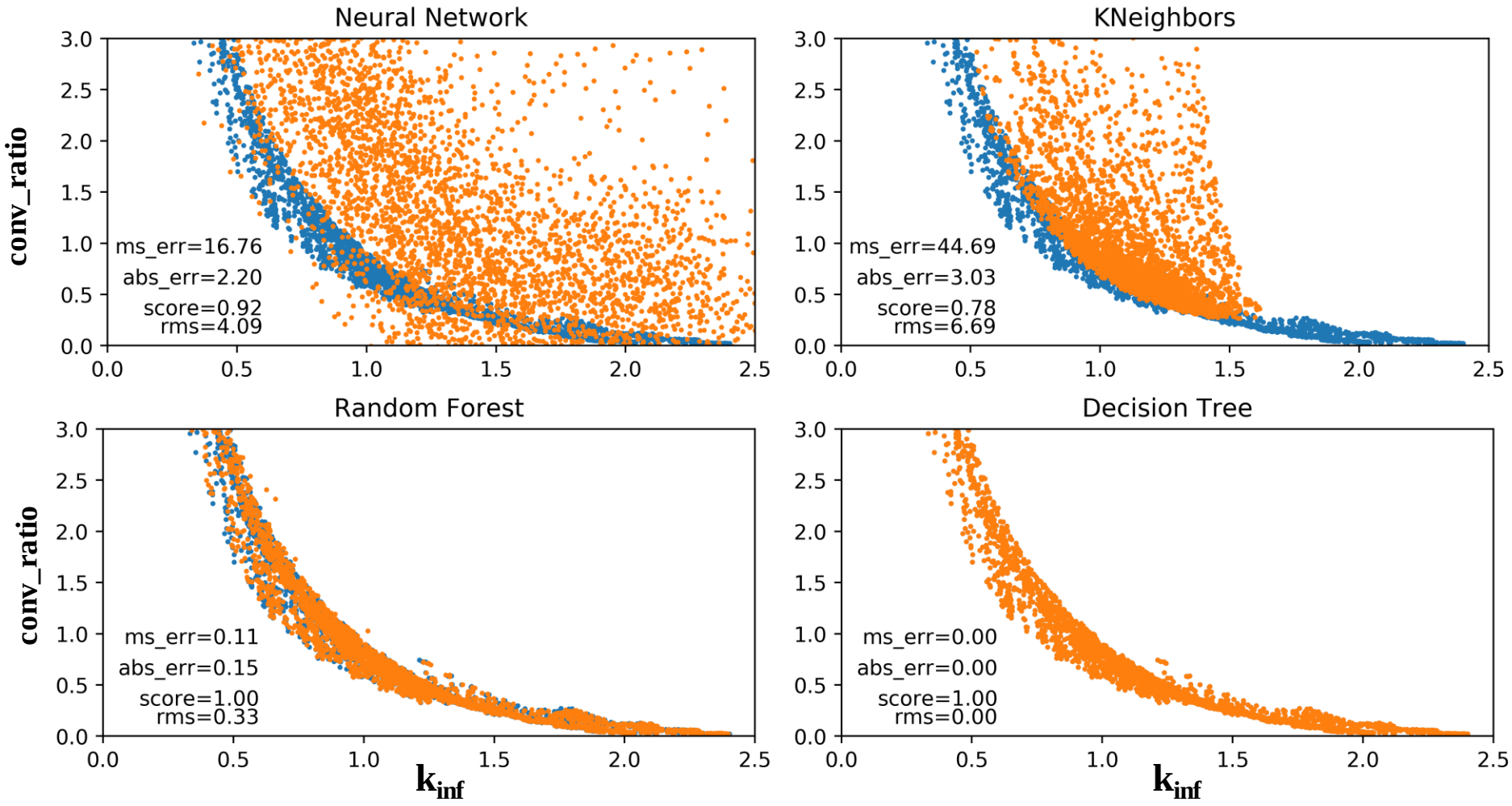






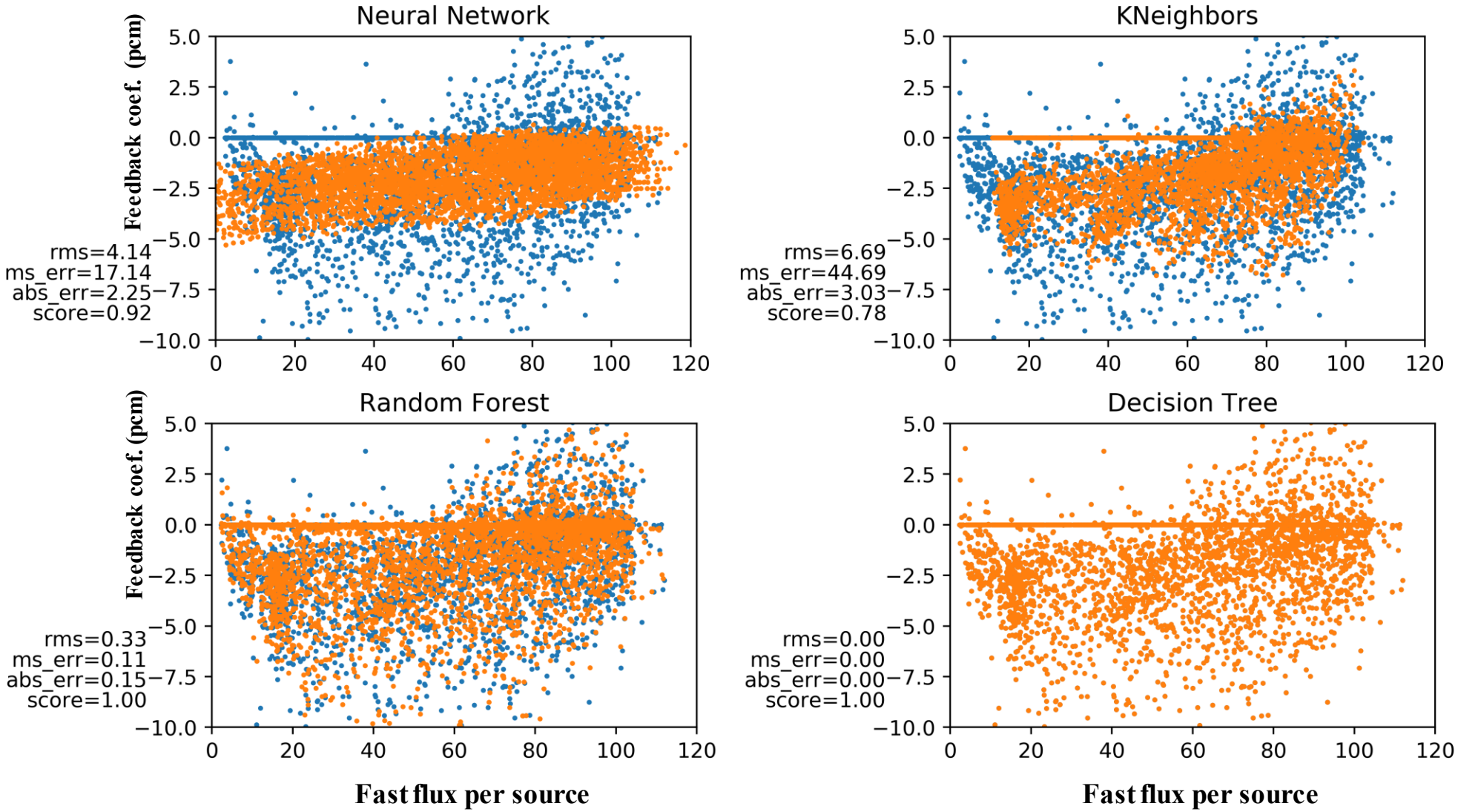


Results: Regressors



- Random Forest and Decision Tree predictions (**orange**) agree well with the real values (**blue**)
- Neural Network and K-Nearest Neighbors need improvement

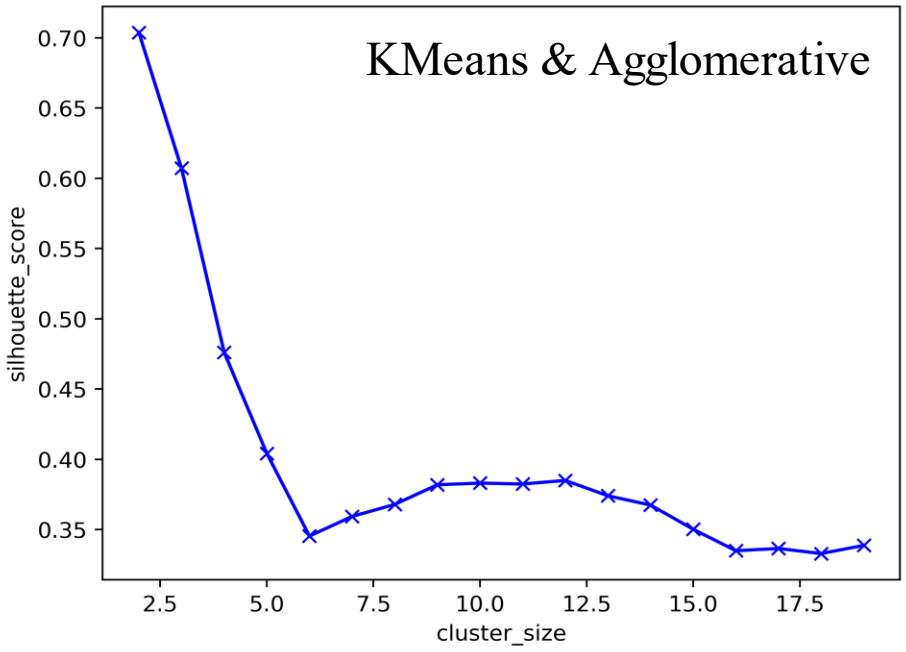
Results: Regressors

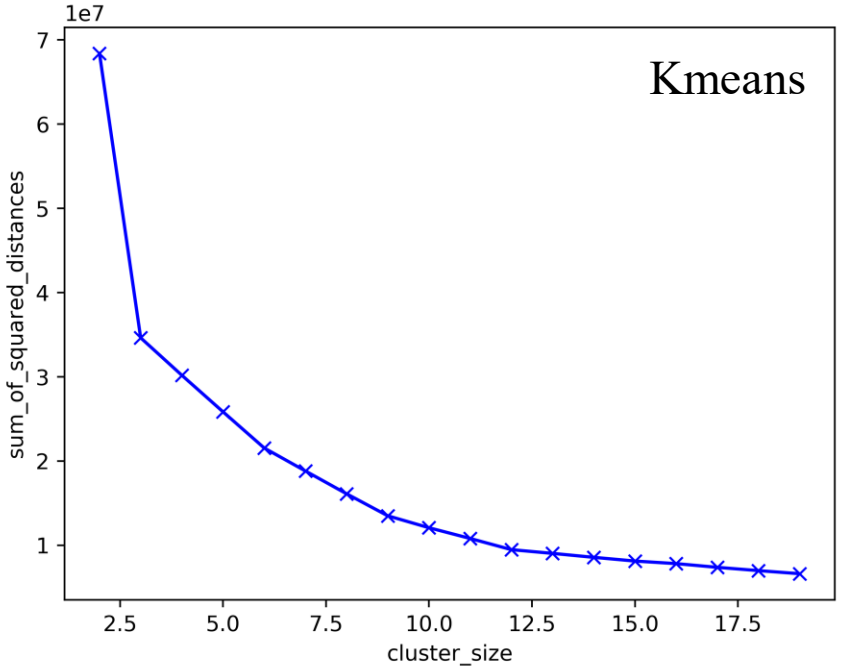


Results: Regressors

- Random Forest and Decision Tree successfully predicted the target values of the test datasets. They have very high score (1) and root mean square (~0) values.
- K-Nearest Neighbors and Neural Network totally fail in training the datasets provided.
- Neural Network yields completely different results.
- Neural Network is highly sensitive to its parameters. They need to be adjusted carefully according to the problem's nature.

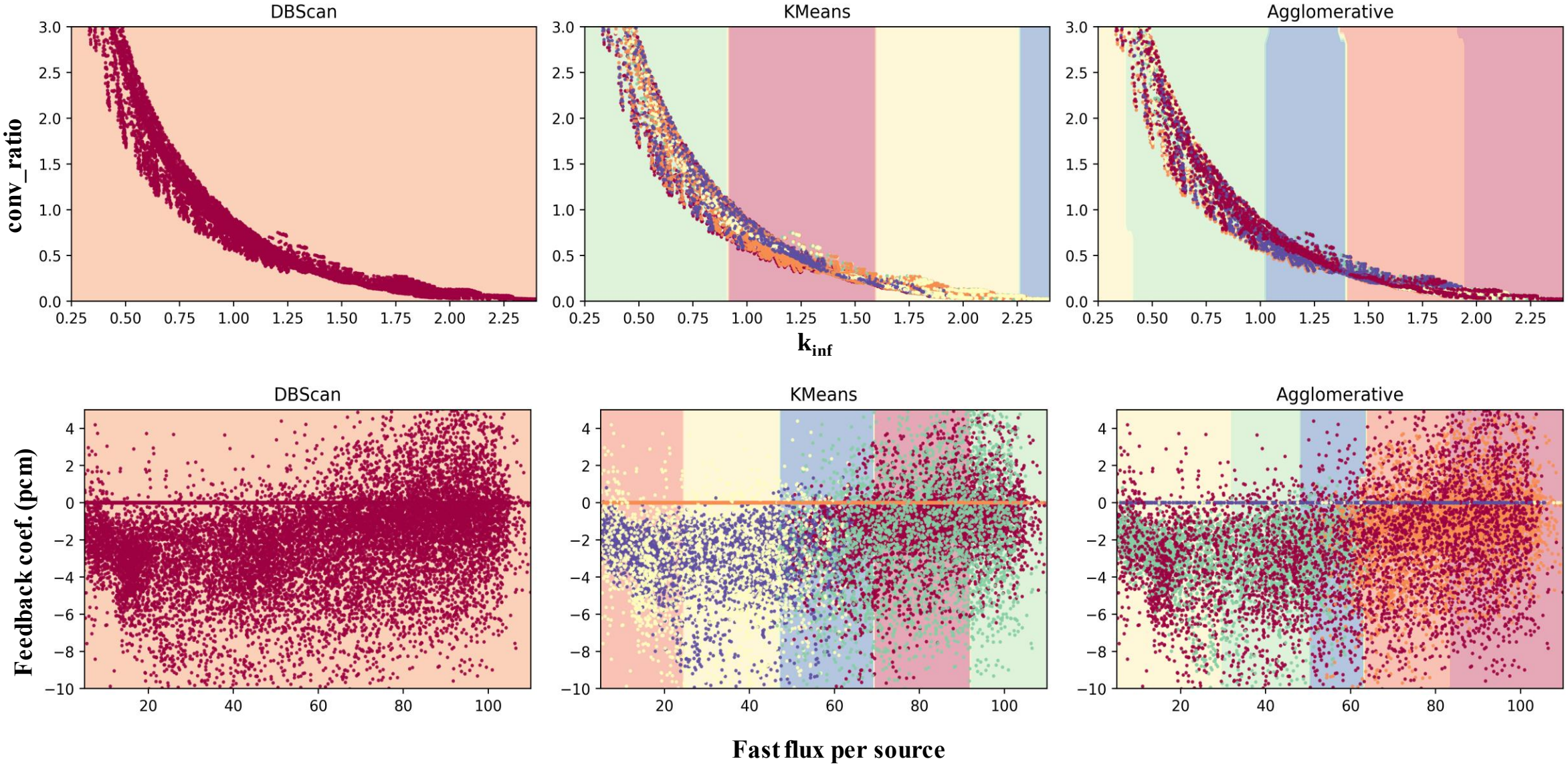
Results: Clusters- Optimal Cluster Size





- Silhouette score and sum of squared distance calculation for cluster size determination
- For clustering, optimum cluster size is 3-5. Set to 5 in calculations

Results: Clusters



Results: Clusters

- Decision boundaries are not sufficiently clear
- DBScan totally fails in clustering the datasets
- Combine clustering method with regression method
- Make cluster, select some classes and finally train the data over the selected classes (as a future work)

Conclusion

- Based on the results of regression analyses, the suggested method gives promising results
- The results show that the method works well for the fuel channel
- It is possible to use this method to predict any target without running reactors physics code
- Selection of the right regressors and classifiers is important
- Categorical distribution in feature and multi-output in the target limit the use of different methods of Regressors and also Classifiers
- Some cluster methods like DBScan do not resolve the search space
- The database used in the optimization seems insufficient as the target space is not well-distributed, needs more data for better prediction
- Some classification methods require class labels
  - Create some labels like ‘fast’ or ‘thermal’ reactor, ‘breeder’ or ‘burner’ (as a future work)

Future Works

- Examine other regressors, clusters and classifiers
- Use Monte Carlo based sampler
- Expand the suggested method to fuel assembly and then to full-core
- Couple to a thermal-hydraulics code (i.e. Moltres)
- Perform analyses for equilibrium cycle with reprocessing option
- Generate temperature-dependent ENDF/B-VIII.0 ACE libs for Serpent with NJOY21
- Burnup and depletion analysis
- Uncertainty analysis

Acknowledgements

- This research is part of the Blue Waters sustained-petascale computing project, which is supported by the National Science Foundation (awards OCI-0725070 and ACI-1238993) and the state of Illinois. Blue Waters is a joint effort of the University of Illinois at Urbana-Champaign and its National Center for Supercomputing Applications.
- The authors would also like to acknowledge financial support from the Scientific and Technological Research Council of Turkey (TUBITAK) BIDEB-2219 Postdoctoral Research Program.
- Prof. Huff is supported by the Nuclear Regulatory Commission Faculty Development Program, the Blue Waters sustained-petascale computing project supported by the National Science Foundation (awards OCI-0725070 and ACI-1238993) and the state of Illinois, the NNSA Office of Defense Nuclear Nonproliferation R&D through the Consortium for Verification Technologies and the Consortium for Nonproliferation Enabling Capabilities (awards DE-NA0002576 and DE-NA0002534), the DOE ARPA-E MEITNER Program (award DE-AR0000983), and the International Institute for Carbon Neutral Energy Research (WPI-I2CNER), sponsored by the Japanese Ministry of Education, Culture, Sports, Science and Technology.

Thank you for your attention

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